Day-one risk coupons: test sheet

Print before the big parts; every coupon uses the real part's geometry functions and numbers. Plates: risk_plate_1_PLA.3mf (20 objects), risk_plate_2_PPA-CF.3mf (26 objects); the fit coupon is in base/pack/fit_coupon. PLA: 0.4 mm nozzle, 0.2 mm layers, 4 walls, 40 % gyroid infill unless noted, 5 top and 5 bottom layers, hole and XY compensation 0, no supports. PPA-CF: hardened steel nozzle (never brass), filament dried (80 C, 8 h) and fed from a dry box, 0.2 mm layers, 6 walls, 60 % gyroid infill, 6 top and 6 bottom layers, hole and XY compensation 0, no supports. The 3MF files carry geometry only: set the material and these settings in Bambu Studio.
To change a number: on the Mac, after bash kedge/setup_mac.sh (from the repo root), cd base and run ../kedge/.venv/bin/python -m tools.risk_coupon (it rewrites base/pack/coupons and base/pack/fit_coupon). Easier: give the failing result to a Claude session and let it change the numbers. Change one number at a time and reprint only that coupon.

#	do	pass	if it fails, change (file: name)
1 Swing-arm direction splitter. Bought: O-ring 20 x 1.5 (arm hub groove), e-clip for the d8 arm pin, 2 x M3x16	O-ring into the arm hub groove. On the plate: sleeve on the centre pin, sun on its rib, arm on the sleeve above the sun with its pin down through the idler, e-clip on the pin, output on its pin, crank on top, stop finger on the column (2 x M3x16). Turn the crank 10 turns each way.	Clockwise seen from above: the arm swings the idler into the output at once and the output turns, no skipped teeth. Anticlockwise: the arm swings out to the finger within about 20 deg of crank, the idler runs free and the output stays still. No jam at engagement.	launch_aim.py: FIT (pin and bore play), arm_solid O-ring groove (drag), A() alpha_d = alpha_e + 20 (swing out), _stop_post_xy; launch_gears.py center_distance extra / spur backlash (mesh).
2 Printed contact switch. Bought: M3x10 SHCS ISO 4762 + M3 jam nut, M3 nut, 2x M3x10 self tapping (pan or SHCS) through the lever root into the pad, at most M3x11.7, 2 wires, multimeter	Nut into the pillar trap (drop of glue), lever clamped with 2 M3 self tappers, contact screw head down through the lever, set the head 0.9 mm above the nut (0.5 mm per turn), lock the jam nut. Meter on beep between screw and nut. Press the lever tip 100 times.	Measure the tip travel with calipers (depth rod from the lever tip down to the base top, note the reading at rest, press until it beeps, read again): beep within 0.8 to 1.4 mm of tip travel (design 1.14), silent when let go, tip stops on the hard stop without cracking, lever back flat, same after 100 presses.	cad/lib/switch.py contact_switch args travel, overtravel, t, b; STRAIN_LIMIT, NUT_PRESS (loose nut), SCREW_LENGTHS.
3 Dog clutch and compliant fork. Bought: M3 nut (yoke tab), 2 x M3x12 (SW_DOG screws in the post), 1 x M3 (post clamp), multimeter	Press the rail into the base. REEL face on the journal (dogs up), collar on the hex (dogs down), fork pads in the collar groove and its sleeve on the rail, dog post clamped on the rail below the fork, SW_DOG screws in the post set 0.5 mm short of the tab nut at full engagement. Push the fork 7 mm down with the dogs tip to tip, then turn the REEL face until they drop in. Turn the face hard both ways, then pull the fork back. Meter on beep between the lower screw and the tab nut.	Yoke flexes when tip to tip (no jam, roofs slide off), dogs drop in within a quarter turn, under hand torque the dogs pull in, never walk out, a light pull on the fork releases them, the yoke springs back flat; the meter beeps only with the dogs fully in, never tip to tip.	_drum_kin.py DEFAULTS dog_backcut, dog_roof, dog_deg, dog_gap, fork_k; drum.py GEOM fork_t, yoke_r, groove, dog_touch, dogpost_s; fits.py SHAFT_IN_BORE (collar on the hex, face on the journal), HOLE_SLIDE (fork on the rail).
4 Face slip clutch with the flexure spring. Bought: 3 x M3x20 SHCS, 3 x M3 nut, string, coins in a small bag, kitchen scale	Stack on the key stub: hub plate (counterbores down), friction disc, pressure plate (pads up), flexure (beam down, tabs on the screws), adjuster plate (traps up); 3 screws from below, nuts on top. Run the nuts down until the flexure just touches (zero), then turn all 3 nuts the same, 1/4 turn (0.125 mm) at a time. Clamp the stub level in a vice, tie a string to a disc lug (r 28.5 mm), lay it over the rim and hang a bag of coins weighed on a kitchen scale. Add or take coins to find the hold and slip weights at each setting.	At each setting the lower weight must not pull the disc round and the higher one must (smooth steady slip, no judder); same result 3 times; the flexure never bottoms before 1 turn. Bands: 1/4 turn: 35 g holds, 60 g slips (design 48 g, 0.013 N m); 1/2 turn: 70 g holds, 120 g slips (design 95 g, 0.027 N m); 3/4 turn: 105 g holds, 180 g slips (design 143 g, 0.040 N m); 1 turn: 140 g holds, 240 g slips (design 190 g, 0.053 N m).	cad/lib/clutch.py: slip_clutch args flex_t, flex_gap, flex_ro, r_in, r_out, n_grooves; MU_PLA_PLA, E_PLA, SIGMA_FLEX_PLA_SUSTAINED (creep: if the torque falls overnight).
5 Bearing seats. Bought: 608ZZ, F693ZZ, 693ZZ, CSK20PP	Press each bearing in square with a flat block or a vice (tile left to right: 608, F693ZZ, 693ZZ); CSK20PP into the round boss with its outer keyway on the printed key.	Firm press, seated flat on the lip, stays in when tapped upside down, spins as freely as before; the CSK20PP key slides in without force, it still freewheels one way and locks the other.	fits.py BRG_RIB_INTERF, BRG_RIB_BORE (all seats); bearings.py CSK20_KEY_RCLEAR, CSK20_OKEY_W (key), KEY_CLEAR in fits.py. If the 608 is off, print tile A of the fit coupon (base/pack/fit_coupon, COUPON.md).
6 M3 holes. Bought: M3x4x5 heat-set insert, M3 nut, 2 x M3x10	Left to right: heat-set the insert (220 C), press the nut in, push a screw through, self-tap a screw in.	Insert flush and does not spin when the screw is tightened; nut presses in and holds; screw drops through the clearance hole; the tap hole cuts a thread and holds a hard pull without splitting.	fits.py M3_INSERT_D, M3_NUT_AF, M3_CLEAR_D, M3_TAP_D (tiles C, D, E of the fit coupon in base/pack/fit_coupon bracket them).
7 Key rib and spigot. Bought: CSK20PP from coupon 5 (inner keyway)	Slide the key stub into the keyed tile and into the CSK20PP inner race (the real interface). Push the spigot into the sleeve slice and turn it.	Stub slides in by hand, key in the keyway, no visible rock. Play: clamp the tile, tape a ruler to the stub end, mark the ruler 60 mm from the axis, rock the stub both ways: the mark moves under 1 mm (about 1 deg). The spigot turns freely; rocked at its free end it moves under 0.5 mm.	fits.py KEY_CLEAR, SHAFT_IN_BORE; shafts.py KEY20_H, SHAFT_PRINT_UNDERSIZE; launch.py SPIGOT_BORE (now 12.35); drum.py GEOM spig_d.
8 Gear meshes m1.25, m1.5, m2, m2.5. Bought: none	Each pair on its plate (label = module), flat faces down; turn or rock through every tooth.	Smooth, no tight spot, a little play when rocked; m2.5: the tall tooth and the cut-down last tooth pass without catching, then the mesh lets go at the gap.	fits.py GEAR_BACKLASH, GEAR_CENTER_EXTRA (m1.25, m1.5, m2); launch_gears.py center_distance extra, spur backlash, sector_gear_2d tall_factor (m2.5); _drum_kin.py pin_x (m1.25 pinion).
9 Shear-pin torque limiter. Bought: string or a strap, luggage scale	Hub slice into the outer slice, pin through both holes (head outside). Clamp the hub lever, hang the luggage scale on the outer lever hole (r 60 mm) at right angles and pull slowly until the pin shears. Break all 3 pins. Torque = kg x 9.81 x 0.06 N m.	Each pin shears cleanly between 5.9 and 10.2 kg on the scale (3.5 to 6 N m, drum_NOTES; design 4.9 N m = 8.3 kg), the 3 pins within 20 percent of each other, both slices undamaged (holes still round); a fresh pin goes in by hand.	drum.py GEOM shear_d (torque goes with d squared), hub_od (radius), shear_s; fits for the hole: drum.coupler shear hole (shear_d + 0.1) and shear_pin (shear_d - 0.04). Spec D8 / D26 limit 5 N m.
10 AS5600 magnet cap (bench test). Bought: 6 x 2.5 mm diametric magnet (comes with the AS5600 board), M3x6, superglue	Glue the magnet into the end pocket (flush, any way round), push the cap on the shaft end (flat to the flat), M3x6 onto the flat. For the bench: AS5600 chip on the shaft axis, 1 to 2 mm from the magnet face.	Cap does not wobble or slip on the shaft; the AS5600 angle follows the shaft smoothly through full turns (no jumps), magnet status OK (not too weak, not too strong) at 1 to 2 mm.	drum.py GEOM cap_s, spig_d; magnet_holder pocket d6.2 x 2.6 (magnet loose or tight); fits.py HOLE_SLIDE (cap on the spigot).